EXP2-1 – The same or different

Comparing data and ensuring that data is comparable!

The **aim** of this task is to develop tools and skills, in order to:

- Present and report data in the correct way (units, significant figures, error, and others)
- Determine whether different samples have the same characteristics or properties, whether they behave differently, and whether these differences are statistically significant

The **objectives** of this exercise are:

- Apply statistical concepts to check and analyze the data
- Critically compare and discuss the results of different samples
- Present a statistics analysis report

Background

For this task, you will work with data from a biomedical engineering student conducting research on targeted drug delivery for treating a rare lung disease. To achieve this, the student planned to use Peptide A to modify the delivery vehicle, which are Polystyrene (PS) particles. The PS particle surfaces contain -COOH groups, giving them an initial negative charge. Peptide A, being neutrally charged, will increase the surface charge of the PS particles after attach to the PS particle at either N-terminal or C-terminal. In order to assess whether the peptide A has successfully attached to the surface of PS particles, Zeta-potential (a measure of surface charge on nanoparticles) is used to characterize the modification effect, a significant increase indicates successful attachment.

The data in the files is the raw data, as it was recorded by the student. However, not all the data may have been collected or presented in the correct format. Therefore, it is also your responsibility to check if there are issues with the data, and decide what to do, if this is the case.

The following concepts will be relevant for this task:

- mean
- median
- mode
- normal distribution
- standard deviation
- standard error
- confidence interval
- coefficient of variation
- t-test
- z-test
- significant figures
- statistically significant
- frequency
- histogram
- binning and bins (in the context of histograms, not in the context of rubbish!)

The **learning outcomes** for this task are to understand and be able to work with the concepts above. In particular how to calculate the relevant statistics shown above using Excel, and how to compare different samples using different statistical methods. Professional skills associated with this task are time and group management, independent study and research, and report writing.

What you should do:

Part 1 Represent the data in a correct way

Group management meeting 1 (pre-session tasks):

Working with your groups:

1. Look at the Introduction and Instructions to make sure you understand the core statistic concepts.

2. Review the Zeta-potential data. There are 4 sheets provided in 1 Excel file; these are selected data for:
   - Unmodified particles from the Student X
   - Unmodified particles from the Student Y
   - Peptide A modified particles from C terminal
   - Peptide A modified particles from N terminal

   Write down comments about the quality of the data presented, if relevant. The data presented were taken as they were provided by student to teachers directly without further treatment. There is likely to be data that may present issues for analysis, as not all students were equally careful about how they gathered, recorded/typed and presented the data.
   Important items to discuss: search for ways the data was presented incorrectly that may affect the results and identify the potential mistakes. Always remember, **data is data, and errors are not mistakes**.

3. Carefully read the instructions for the GMM2 & GMM3 tasks below. Then, prepare a **new Excel spreadsheet** *(you can find the **template** in 'Raw data file' folder)* with ***available*** Zeta Potential data (round with **significant figures**) that you will need to analyse for this exercise. This Excel spreadsheet will be submitted together with your report at the end of the task ***as supplementary***, including new sets of data, all calculations processes (functions should be kept) and graphs that you have carried out. *You also need to include the essential graphs in the main context of your report.*

Group management meeting 2 (please be prepared, you may initiate work ahead of the day):

Working in your groups, consider the following discussion/questions, for which you will need to make calculations using Excel spreadsheets and start writing a report.

Section 1: A discussion of the raw data as presented (see GMM1 tasks), including identification of data that presented serious **issues/mistakes** (not errors), and what was your decision about it, including a short justification.

(No matter if you decide to delete, change or keep the data, you need to show your justification/evidence.)

Section 2: What parameter(s) provides you confidence in the zeta-potential for each particle from the series of measurement? In other words, how would you judge the quality of the results in the zeta-potential (generated from 30 cycles) for each particle.

Section 3: Please round all the available zeta-potential values with significant figures and calculate the mean, mode, median, SD, SE, CI and CoV for all the samples separately, and express the final results with **significant figures**. What are the units of the values in the raw data files? And what would be the best unit to use for results representation and why?

Section 4: How are you going to describe the unmodified samples (from both Student X & Y): do these samples follow a normal distribution? Can you investigate this in a graphic manner?
Note: Do not focus on doing further research, it is nice of you doing this but has to be beyond your critical analysis based on what we have taught in class.

Part 2 Significance tests

Group management meeting 3 (please be prepared, you may initiate work ahead of the day):

Working in your groups, consider the following discussion/questions, for which you will need to make calculations using Excel spreadsheets and complete writing the report:

Section 5: Are the data of unmodified particles from the Student X and Student Y comparable?

Section 6: Are the data from the C terminal-modified particles and N terminal-modified particles comparable?

Section 7: Are the data between unmodified particles and modified particles comparable?

For the 3 sections above: *If they are comparable, at least two methods should be employed, one of the methods should be in graphic manner; if they are not comparable, please analyse and explain why, and you are not expected to use other more complex statistics analysis tools/methods out of the session at this stage.*

Section 8: A brief discussion of what you have learned and how you dealt with problem solving and group management (corelate with PES requirements).
Nope: Do not just simply describe what you have done, try to include more meaningful reflections.

Group management meeting 4 (post-session tasks):

Finish the report containing a selection of essential graphs that support your discussion in GMM3, further graphs can be appended on the Excel spreadsheet. (*Your report must be a complete report even without Excel spreadsheet as supplementary) You will submit this report, together with only **1 Excel file** that contains all your relevant calculations and graphs. (Do not re-submit the original Excel spreadsheets that you have been provided on QMPlus, in total, or partially, just an Excel spreadsheet that shows your work in this task).

The report will be **no more than 1500 words** (exclude references) containing **8 sections**. You can give subtitles of each section as you wish, but make sure the structure of your report follows a reasonable logic order. You may use narrow page margins, letters no smaller than font size 11 and line spacing no less than 1.15.

The **title** of the report will be: **EXP2-1 Same or different Report XX** (where XX is your group number, e.g. M16 or P8). The names of **all the group members** should be given as authors of the report *(the group member who doesn't appear as author might be considered as poor contribution with further investigation).*

You will **submit** your group's report as a **pdf document on QMPlus** using the file name:

'QXU5017 EXP2-1 Report XX' (where XX is your group number, e.g. M9 or P17).

You will **submit** your new Excel file together with your report **on QMPlus** using the file name:

'QXU5017 EXP2-1 Data XX' (where XX is your group number, e.g. M9 or P17).

The report must be completed and **submitted by 24:00 on Wednesday 5th March 2025**.

The report must be the **work of the whole group** and the content agreed by all group members.

Only one group member needs to hit the submit button to submit the report.

Marks breakdown:

EXP2-1 is 25% of the marks for QXU5017

Schedule:

M Groups

Date:	Time:	Place:	Activity:
Monday 17th	10:30-12:10	QMES A310	Introduction to QXU5017 & EXP2-1 Part1
Wednesday 19th	10:30-12:10		Group Meeting 1 (Review and check the raw data)
Friday 21st	08:30-10:10		Group Meeting 2 (Write Report Section 1-4)
Monday 24th	10:30-12:10	QMES A310	Introduction to EXP2-1 Part2
Wednesday2 6th	10:30-12:10		Group Meeting 3 (Different sample comparison)
Friday 28th	08:30-10:10		Group Meeting 4 (Complete report Section 5-8)
Wednesday 5th	10:30-12:10	A310	Report Submission

P Groups

Date:	Time:	Place:	Activity:
Monday 17th	08:30-10:10	QMES A310	Introduction to QXU5017 & EXP2-1 Part1
Wednesday 19th	08:30-10:10		Group Meeting 1 (Review and check the raw data)
Friday 21st	10:30-12:10		Group Meeting 2 (Write Report Section 1-4)
Monday 24th	08:30-10:10	QMES A310	Introduction to EXP2-1 Part2
Wednesday2 6th	08:30-10:10		Group Meeting 3 (Different sample comparison)
Friday 28th	10:30-12:10		Group Meeting 4 (Complete report Section 5-8)
Wednesday 5th	08:30-10:10	A310	Report submission